ActInf Livestream #031.2 ~ "Non-equilibrium Thermodynamics and the Free Energy Principle in Biology"

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hello everybody and welcome to the active inference lab today it's october 26th 2021 and we're in active live stream number 31.2 welcome to the actimf lab we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links here on this slide this is a recorded an archived live stream so please provide us with feedback so that we can improve our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams we're here in our second discussion group discussion for 31 towards the end of october so also happy halloween and this is our jumping off point to many ideas so looking forward to this cool discussion the goal is going to be to learn and discuss this paper by patricia palacios and matteo colombo both of whom are with us today so thanks so much to the authors for joining and feel free to address what you can while you're here and then just leave whenever works for you and in this discussion we can take it many different ways because so many topics came up in the dot zero and dot one but we have all of the formalisms some big questions a lot of topics that we raised in the 31.1 video as well so we can begin with an introduction and warm-up section we'll have the non-authors introduce themselves and then just say hello and then we'll pass to the two authors both of whom are joining for the first time to a stream so again thanks for joining and we look forward to the discussion so i'm daniel i'm a researcher in california and to just add something i'm excited about matteo and i were just discussing right before we started some of the sociology meets history and philosophy of science and oh yeah who could forget about the actual topics themselves so there's just some cool intersections that we'll be able to talk about i'll pass it to dave i'm retired from information technology i'm in the cool dry mountains of the northern philippines and now that i don't have to work for a living i'm getting really busy on learning cybernetics which is when one of the things i did back in college and having a great time is fired up about having a complexity weekend ramp up starting tomorrow sojourn stephen we can't hear you steven but you're unmuted so maybe your audio selection is odd but let's um welcome steven back so figure it out and let's go to um patricia first welcome and happy to hear your introduction hello well thank you for for the discussion for for the interest in discussing our paper so i'm an assistant professor in salzburg i'm originally from chile and i work in the philosophy of physics philosophy of science and now i'm working more and more in the philosophy of the complexity science and i have done some work on going to physics and well i'm interested in general in the in the statistical mechanical applications to other disciplines um well statistical mechanical models in economics biology and well chemistry although i haven't explored that already yet i'm interested on that with mateo was my i did my first exploration in the free energy principle but um just to be clear i'm very interested in the application of statistical mechanics to other disciplines even if we appear to be critical to that in this in this paper thank you matteo thank you very much for uh the kind invitation and also for the spotlight uh on our paper completely undeserved my name is matteo colombo i'm based in tilburg in the netherlands uh as my accent might suggest i'm italian i'm originally from milan and mostly my work lies in the philosophy of cognitive science i have a special interest in computational modeling especially reinforcement learning and bayesian approaches uh to brain function you know it's been fun to collaborate with patricia at this project which um you know combined a bunch of interest already had and helped me to learn something new great well maybe we could just start with some of the context of the paper like how did both of you intersect towards this very interesting view on fep and active so were you studying active inference and then wanted to integrate it with statistical physics or vice versa i think we heard some of the seeds of that but it'd be good to clarify yeah if you want yes okay so i think patricia and i started to chat about this uh two or three years ago at a conference uh problem uh so my background in uh is in uh quantum neuroscience and uh philosophy mainly so masters involved and during my work at edinboro as a phd student i was actually co-supervised by a person peggy serrier who is an expert in psychophysics and then she moved slowly uh to apply tools from basian modeling and reinforcement learning modeling to psychiatry and so my and my other supervisor was andy clark and so was a good um say uh environment for me you know to learn about active inference i think those were the early days of this approach and started to are asking questions basically about the approach in terms of actual applications a very limited experience i suppose that the only uh serious modeling experience said was during my uh phd where with another collaborator we started to model uh things in uh abroad bayesian framework um this paper came in to be i think mainly because of my own interest in bayesian modeling and patricia's own interest which she just mentioned in statistical physics approaches to economics and to other sciences and so we started to ask hey in terms of scientific representation that's the focus of our paper focus is not so much as also in written in the paper on the metaphysics of you know active inference but more about japanese language this formalism the scientific representation and let's ask what kinds of opportunities and challenges that this formalism presents when it comes to applying it outside its say original field as in you know statistics statistical physics and so we started to chat and uh with a back and forth for a while and then we say started to put into focus various opportunities and challenges in applying concepts from thermodynamics and statistical physics to make sense and understand organisms and life science phenomena and what you know we found very interesting is this kind of trade-off between the generality of it that is it seems that the machinery of the free energy principle or active inference uh you know we might have a discussion about the difference if any between the two how this machinery allows allows you to cover at least a model many types of systems at least allegedly but with a sacrifice in a plausibility uh what we call a realism in the paper and again by itself is trade-off we don't think there is anything wrong with uh you know going for maximal generality in an approach that might cover you know or at least allegedly from cells up to you know cultural phenomena um and that's our paper in a nutshell that is how these trade-offs might depend on a choice of language or formalism to make sense of allegedly any thing and again the thing good is one of the key terms i think that the maxwell and you know before him carfreestone uh put on the map here and i think it's a helpful way also to look at our contribution again in a nutshell i think the framework of thinggood uh might miss out the most interesting properties of the kind of thing if there are things um say organisms happen to be thanks for that patricia anything to add i'll suggest just um to to repeat what what matteo said so my interest in the free energy principle came from my interest in the application of statistical mechanics to all the disciplines so that's why we started discussing this and that's why i got so excited about this about this research and i agree with the summary of the paper that mateo just just did this thread of grounding in fundamental physics ideas and then trade-offs or opportunities and challenges really became clear so maybe one general opening question what does it mean for a non-statistical physics idea to be grounded in statistical physics like we're intrinsically talking about some interdisciplinary or transdisciplinary or applied scenario so we're leaving that thing that field of statistical what baggage good and bad do we take with us and then is it an advantage is it a weakness what does it actually mean for a non-physics area to be grounded in yeah thank you for the questions that's one of the questions i have motivated my research in the last years uh i don't think there is a unique answer to that so the more i do research on the applications of statistical mechanics to other discipline the more i understand that it varies so it depends on how it is used how how statistical mechanics is used and it depends also on the case studies and disciplines and the specific models so i think that's why it's worth to look at specific case studies to specific applications of free of statistical mechanics to the disciplines in order to answer that question like on a case by case basis i think in some cases the application of statistical mechanics is well justified especially when the idealizations are well justified in some other cases as i would say tend to think that it's the case of the free energy principle the application of statistical mechanistic lack justification so we are not saying or we are not concluding in the you know paper that that cannot be i know that it doesn't make sense to propose a principle that is based on statistical mechanics but we just think that there is much more work to be done in the justification of especially of the idealizations that are involved when we use statistical mechanics to formulate a principle like the free energy principle classic philosophy where do we air or miss when we idealize make ideas of or generalize specific systems or things and what is the thing concept or the system concept in either patricia or mateo in either of your areas like if you mentioned a thing or a system what would that mean in to the the disciplines that you're working with and then how is it being used differently in fep and active and then i see steve my dad you want to answer this sure um i mean in a sense your question is about metaphysics uh and the question is what's the thing and what uh in what way we can make sense of a thing good uh with you know a formalism like the formalism markov blankets for instance or the formalism of uh statistical mechanics just to turn the point slightly in a different direction if you may i suppose the uh question about thing good and the role of idolizations on which you know patricia just commented on is not just boiling down to the to the you know conclusion that ah that's difficult let's move on uh to to apply these tools um to a certain system uh the challenge we put on the table here is concerning the generality of it uh the generality of the act inference uh you know there is a excellent work uh by uh philipp schwantenberg and others using act inference models uh in uh market decision processes with humans you know fitting decisions in reinforcement learning or decision tasks with active inference models and again this is just one example of a productive use of the active inference approach which you know resonates with regular approaches like bayesian modeling or reinforcement learning modeling uh we're taking issue with is look you're making claims about a maximum scope okay maximum scope kind of application of this approach and to sensibly apply you know models at any scale you know any scale even outside the life sciences you know having to do with economies or thing good certain assumptions must be in place and these assumptions you know we provide a number of examples which all involve the availability of large data sets okay that can constrain uh say the application of the model and also background knowledge about the target this is the target system and the thing you want to understand and especially you know bridges conceptual bridges between parameters within your model and features of the actual systems you want to represent so these are the three i think challenges that we put on the map for again the large scope or very general scope some understanding of active inference or the free energy principle so this is a bit of topic so i didn't really answer your question about thing but at least these are motivations we had to write this paper as in making claims about maximal scope come commitments and these commitments at least in the case of many or most biological systems do not look uh satisfied or plausible thank you in a way it does oh wait maybe a little bit mateo but um in a way it does address this question about our models as things and their relationship to other things perhaps the target of modeling so patricia if you want to add anything otherwise we can go to stephen just to stress some of the things that matteo said because i hear the session offline i hear the session of last week and i i i saw that um maxwell was concerned or pointing out that we didn't take the cases that the applications of the free energy principle to some systems in biology seriously and but the thing is that our claim we did take them seriously and we mentioned them in the paper but that doesn't contradict uh our claim because our claim wasn't that the statistical mechanics cannot or the language of dynamical systems cannot be used to model biological systems our claim was about the generality of this of this principle so basically we were concerned or where we were casting doubt on the possibility of formulating a general principle like the free energy principle that presupposes that all relevant biological systems can be modeled as random dynamical systems so just to emphasize that our claim was our concerns with the free energy principle had to do with generality of this principle and not with the possibility of modeling uh some biological systems in that way using the language of random dynamical system thank you so stephen and then anyone in the live chat or anyone with a raised hand here yeah thanks this is very interesting hopefully you can hear me okay um i'm curious um i know that in the the the reference you make to the paper that is um life as we know it so there there's that question about um suggesting that life or biological self-organization is an inevitable and emerging property of any ergodic random dynamical system that possesses a markov blanket and i know that in some ways i i i sense that what they've done to try and deflate that claim from 2013 is maybe to go down the mathematical route i s i sense that you're saying they need to maybe also go down the physics route would that would that be something that you would say like in terms of how that um that statement which has to some extent been couched over the last few years um are you saying that these you know the the physics aspects um haven't been addressed as much as the statistical aspects i can say a couple of things and then i leave it to you patricia yes very very quickly um so um one uh line uh of criticism about our paper is that you know [Music] the active inference approach has been evolving over time and so we're making claims about ergodicity and which are just obsolete now because uh you know from the coast hostile paper uh that assumption seems to be dropped uh and replaced with uh the the assumption of stationarity but then the question actually two questions one question is why should we believe that any uh life science phenomenon any organism is stationary this what is it what is the justification for that assumption and the second uh observation before i leave it to patricia is that even within the paper you know that the constitutional paper which is you know theoretically uh very rich they frame the contribution to biology in terms of a metaphor explicitly early on they're saying look what we're seeing here is just a metaphorical uh representation and so in a sense that's what we're being arguing for that is sure you can gain this maximal generality but at the cost of uh making assumptions either ergodicity or structurality now which might might not apply to biological thank you patricia yeah so to to answer stephen's concern i would say yes so one of the things that we wanted to point out is that we need to focus on the on the physical foundations of the free energy principle a little bit more than only the statistical foundations of the free energy principle so we believe we don't claim to be the only ones that focus on on that but we believe that there's lack of literature in that sense and we wanted to contribute to uh to to that to fill that gap what we call feel that god uh so that's that's the short answer and then um about again the discussion that you had last last week uh you saw maxwell i think as well was complaining that we focus too much on the older formulations of the free energy principle instead of the enumerations and he basically said that ergodicity doesn't play any role it's not even mentioned in the new formulations of the free energy principle i don't have a biological background or background in neuroscience and i haven't worked for a long time in the free energy principle but i do have a background in physics and i am a little bit puzzled with the idea that ecbodicity doesn't play any role in the new formulations of the free energy principle especially because they explicitly allude to the existence of weakly mixing random dynamical systems and to my knowledge mixing systems require erodicity so implied erodicity so i'm really not sure and i haven't seen explicitly how can you drop the assumption of eliciting the free energy principle so i think that if they if some authors working on the free energy principle could clarify to what extent or why you don't need their basicity anymore i think that would be helpful so we would be grateful for that we wouldn't consider that as a failure in our project so we would consider it as a success so if our paper leads to more clarity in the free energy literature that would be a successful paper for us and and i think the fact that bill edal also some paper that was quoted in this discussion last week also allude to the older formulations of the free energy principle just shows maybe a confusion in the literature between older formulations and new formulations so i think again it would be very helpful for people interested in this at least to to have more clarity about what has changed in this new framework in these new formulations of the free energy principle and to have a more consistent framework that one can work with because we were interested in working with the legitimate framework we weren't against the free energy principle and then tried to look for the weak for the weakest argument to attack them so that was never our strategy i don't have anything but i am not married against the free energy principle i don't have anything a priori against crystal so the same with matteo so we are just generally interested in in this application because it can be actually potentially useful or because it has been so uh it has caught so much attention in the in biology so that's why we are interested in this in the first place not with the idea of attacking the free energy principle so again if we could have better understanding of of why the new formulations don't require ludicity or about what are the key assumptions in the new formulations of the free energy principle i think that would save us time and we would be happy to to look at that so i i think again i i agree that we focus mostly on the older versions of the free energy principle but i think it was partially because the papers were clear and not because we wanted to attack the weaker papers in the same like the fact that they appeal again to the existence of weakly mixing random dynamical systems suggested for me that they still are using ergodicity and apart from that ergodicity was not the only argument that we were presenting in the paper so we also mentioned phase spaces and attractors and the new formulations still require phase spaces because they need some probability and they uh they also consider or model homo aesthetics or non-equilibrium steady states as uh in terms of attractors and that again falls into our criticism because we were pointing out independently of the ergodicity assumption we were pointing out that it's very hard to model in most systems biological system even physical system perhaps um stationary non-equilibrium stationary states in terms of attractors because attractors implies some asymptotic results and it's not easy to get these asymptotic results and so basically that was our criticism so we were by no means only focusing on eroticity even if ebodicity doesn't play any role in nervous formulations in in new formulations which is not clear to me but even so i would say the other points still remain and we would like to to have a clear answer to that even even if that means that we have to i don't know say that we were wrong in the end but um i insist so if our paper leads to more clarity in the literature that we will be fine we will consider it as success thanks for that and welcome blue so i'll just give one comment and then stephen so absolutely about the evolution and changes it's a non-stationary or maybe there's a higher level of a stationarity to the literature but it's one reason why blue and i and maybe you if anyone listening wants to contribute seriously to this are very interested in tracking nomenclature and which letters are used to refer to internal and external states and which assumptions are implicit or explicit how does that play out with the evolution of literature in this area so that's like something that we'll hopefully will be able to point to maybe it's going to be version 1.4 in this paper or you know version 1.4 the the english gambit we can name them like chess openings okay stephen yeah i think you make a good point i mean i think also i i do agree in the de costa paper stationarity has really given us another way in to some sort of state that gets revisited it doesn't necessarily take out a good city maybe adds another potential so there's this this question doesn't it can't just be answered from a math perspective like i think it's there's this physics um reality and maybe this is where non-equilibrium chemistry starts to come in or that's a new field um and i i think that your your points are are useful in clarifying some of those questions i suppose the question that also comes up which is maybe less common in physics is how much can science how much can um a kind of noisy version of ergodicity be enough and at what point you know how many beers is it before i fall over you know at what point does um does the active influences process come down and i i don't actually think that's necessarily been brought in but i don't know if physics knows quite how to handle that question yes how stringently do specific realizations or models especially if they're being evaluated on utility how stringently must they fulfill some of the underlying frameworks like we talked about how the t-test is built around equal variances unless you're using an inequal variance version and of course those will never be precisely realized but then how close to being precisely realized must or should things be um matteo or patricia any thoughts oh otherwise the the comment that came to mind was patricia you mentioned um and it was in your 2018 paper i think that we looked at in the dot zero about how there's the asymptotic characteristics and the convergence time so that was very interesting um just that there was a there was it's sort of this attractor but there was more to the attractor than just where it was so maybe unpack that and then how did you see that grafting into or relating to the um comments on fvp and active yeah exactly sorry i have to leave now thank you you know it's been fun uh i'm very sorry you know it lasted only 30 minutes because i saw so blue joined and i had answers to a couple of their questions in my previous sessions about dynamic equilibrium and why we use internal uh states uh as active states well we can do 31.3 literally anytime we can do it in a year so that's why we do decimal notation any time it's right just let us know and you're always welcome back for a guest stream presenting on current work so good luck with your teaching responsibilities and see you soon too generously church hello everybody and thanks for paying attention cheers to patricia okay yeah you were mentioning my 2018 paper thank you er yeah so that i think the the question of asymptotics is related to the question of approximation science and i am interested in both questions and this is basically what i have done independently of the free energy principle in the in my research in the last in the last year so basically what i what i pointed out about asymptotics in the paper that you were mentioning is that um in order to make a model realistic a model that takes a synthetic into account in order to make that model realistic we not only have to pay attention to the solutions that you get in the limit when you take a limit you also have to pay attention to how fast that limit converges in that time that can be considered relaxation time or that time that it takes that that the limits needs in order to to or that we need in order to achieve that limit um needs to be realistic too so you don't only you you don't only have to pay attention to the values that you take that you that you obtain when you take a limit you also have to to pay attention to the to the rate in which you approach that limit so that will also help you to understand how realistic is the model that you are that you are using how would that tie to approximation science uh it depends again some some models like on the models that you're analyzing in the science that you are looking at so if you are investigating models that use limits then there is a very clear understanding of approximation because uh limits give you sort of a neat understanding of the approximations that you are using but approximations are not always uh in the form of limits sometimes approximations can be in the form of similarity and then saying to what extent that model is realistic it can be a little bit more obscure so when your model implies limit then you might have a clear way of understanding to what extent your model is close to to the to the target that you are trying to mode very interesting so thought that's um come up in a few different contexts is um it's almost like a vitalism 1.0 and a vitalism 2.0 and the 1.0 was like you know is there the elan vital is there something more to it than chemistry in biology and you know the synthesis ex vitro or x vivo should i say of vitamin c for some reason it's pointed to as an example of like demonstrating that biological molecules could arise through chemical synthesis and therefore maybe more than just vitamin c but maybe dna could also be made in the test tube and now we're in a area where we're asking okay given that grounding of biology in physics and chemistry 1.0 how do we go from that sort of equilibrium into homeostasis and allostasis and more advanced forms of predictive higher level behavior from systems that are also composed of less strategic components there's going to be just like there were challenges going from the physics of a membrane into understanding how biological systems deploy membranes now we're moving up a level into the cybernetic or into the anticipatory space and asking how those very same biological systems which we've deflated away to be chemical and physical now will there be another level of deflation or is this a bubble that's a little harder to pop steven and then patricia or anyone else with thoughts yeah this is a good point about it being a bit harder to pop um because there's a lot of uh once you get into the chemistry there's a lot of there's a lot of hidden states let's put it like that so there's that assumption that you pass through often maybe a transition state when you go from reactants to products but it's going from equilibrium to equilibrium even the transition state is looked at in terms of its energy accessibility but that's again a kind of a stable state so this question about something circling around non-equilibrium or far from equilibrium um mixing states which is why i think you can't i think they say ergodicity can't be or something along those lines can't be ignored uh certainly once you get into anything which is more in the liquid uh contexts um so the question comes up then is when you've got these attractors or these far from equilibrium states which are not necessarily energetically favored alone they may be somehow entropically favored or somehow complexity favored through these attractors you know is that something around how catalysts are working um in the more sort of if you were to go in at a more detailed level of modelling um so i wonder how your thoughts are around the sort of catalysis processes that might be happening in biological systems and how you know how do these far from equilibrium states facilitate these these abilities to minimize free energy if you have any thoughts not patricia otherwise blue do you wanna i would leave that to blue we have to work more on the free energy principle i don't know i i think uh the assumption that i've worked more on the free energy principle is maybe kind of i don't know not not right i don't really have an answer don't let the cat ears deceive you [Laughter] cool questions um there's many other avenues we can go but patricia what is your current research or what do you see following your work now that it came out like what where do you go from that state update to your next questions if they're related or not it's all good yeah actually i'm thinking about applying for a project uh which it doesn't have to do with free energy principle but it does have to do with applications of statistical mechanics to biology so i have been working for a few years i will probably i'm so i just um finish a book on emergency reduction in physics so i have been working on a concept of emergency reduction for physical systems especially thinking about critical phenomena and what i want to do now is to see whether that notion of emergence and reduction that i suggest also works for biological systems but i will focus on biological systems that can be modeled as critical phenomena which means can be model using the physics of phase transitions so this is what i what i have in mind now and i i i intuitively think so my binary uh hypothesis is that the same concept doesn't work for biological systems and well that one needs to distinguish between them but i think it would be useful to look at specifically at systems that can be modeled as critical phenomena to see this analogy maybe between biological systems and physical systems so that's maybe the reason why i will focus especially on those systems that can be according to to to many physicists b model and biologists be modeled as a critical phenomenon interesting in fact criticality is uh perhaps again the language analysis will reveal in the future but it's not a term that gets brought up in the active and fep area that much it's more apt to describe assist although in fristen's work on neural systems there actually is a lot of it and so there's actually so much in the spm and dynamical modeling literature that has not made it quite over the hill but i think that there's more to be said we'll look forward to that book stephen yeah just that's an interesting point and they're making um can i just ask so is it almost like you think there's a lot of a lot of computational processes that can happen purely through phase transitions and statistical processes which is not quite the same as an inference process but it is it may be more like the sort of thing you see when molds um find their way to food and things like that so you sort of saying there's a kind of a physics explanation for some things which doesn't require an influence process and that could do a lot of the heavy lifting yeah so i'm i will focus on on on models that use the mathematics of phase transitions with variations so you never use exactly the same mathematics you have to you have to make change depending on the case study that you're investigating but i will i will focus on on models that um that use the mathematics of phase transitions to understand biological phenomena an example would be **[ ] behavior for example so there are plenty of models that use the icing model or something like the icing model uh to understand falcon behavior which is that behavior of birds that seem to be emerging with respect to the behavior of individual birds and in** other case studies like melding protein melding that can be modeled as a phase transition so i'm going to i'm going to analyze plenty of case studies of research that have been done already in physics or in biology and the idea would be to understand what kind of notion of emergence do you find in biological systems and the question is whether that is similar to the notion of emergence that appears in in physical systems and as i said i intuitively think that it's not the same but since i will look at similar case studies i think that can show the difference between biological systems and physical systems quite quite clearly there's one thing i was curious then is it like because this is interesting when i'm talking to people who've got maybe more of a chemistry background emergence is more the emergence of interactions between things right molecules and stuff and then you get these kind of and then you've got this more process based kind of inference dynamical one so are you you sort of seen the way of emergence being able to do quite a lot from the kind of perspective of things interacting with each other based on rules rather than non-linear dynamical both actually so you find both in phase transitions that way that's why phase transitions in physics have been considered as the prototype example of emergent behavior because you find non-linearity you find um long range correlations and well the interactions in most systems are local but you still have these long-range correlations that require using some fun segments like the randomization group methods for example so you have some notion of collectivity there in in phase transitions in physics not because of long-range interactions but because of long-range correlations so it appears as if the interaction between between uh distant particles would be would be long-range it is not but it appears like that because you have long-range correlations so i will look at similar cases in biology that have been modeled already using physics and then i will try to answer whether that notion of emergence in terms of non-linearity and so on what i suggest it's emergence emergent behavior in physics can also be applied to biology and i think well but this is this is preliminary so this is not these are not the results of my research because i will be doing that for the next years but a preliminary thought you will see a difference especially with top-down causation that might happen in biology that doesn't happen in physics but this is just preliminary so again we try not to say too much about this because i haven't done the pro i just have done the research for the project but i haven't done the project you can you can wait for this result blue so that's super interesting um and again something that like i'm very interested in is this um idea of emergence and when you're you're studying phase transitions or criticality in physics

like for example just to from a liquid to a gas you know it's easy to see on a graph this kind of inflection point of temperature where the phase transition occurs and i just am wondering what kind of metrics you're using to quantify emergence um in your future research in biological systems uh that depends again sorry that many many times my is the same but that depends on the i'm not using a specific metric because i'm not proposing as a general framework i'm just studying like different case studies and depending on the case study on the kind of phase transition that i'm studying is the the metric that is attached to it but i'm not i'm not suggesting a general formal framework that has a metric in itself so i'm i'm studying like case by case so can you maybe provide some examples or you don't want to say yet too much in biology yeah the [ ] behavior so there are different different models there i can you can you can send me an email if you want and i can address you to some this literature but there are plenty of models of funking behavior right that use that use models that are already so that sort of migrate models from physics so there's some of them use a slightly revised version of the of the icing model it makes me think about how uh the sort of initial target system let's just say a solid to a liquid phase transition and then there also are potentially critical dynamics related to information and decision making which comes into play in active inference where you could have uh a uh non-linear relationship between information coming in and then it's in policy mode a and then there's a phase transition to policy b and so um it's kind of like again that vitalism 1.0 2.0 there's going to be a lower bar a phase transition to talk about well the protein folding in the cell is going to be a special surrounding that facilitates it to do special kind of transitions physically but then how do we think about higher order behavioral properties especially when they exhibit some tantalizing similarity or analogy or metaphor like we've been discussing but can we just take those same models and think that it's going to be as clean all the way up um blue and then stephen so in the flocking model um for example like and you're talking about downward causation uh and something as something that you wish to study and so i'm sure that you've maybe read um the information theory of individuality paper by um david pakkauer and that group at the santa fe institute so there's um a point in which you know this like group of birds that's like flying around you know kind of chaotically um assembles and forms like a flock and and um flies together in in like a kind of patterned way and so like i'm interested in in terms of criticality at what point is does downward causation occur right and and is that the point that then it stops being like the the ensemble becomes a group a unit that's functional like you know because there's it's like that's a good kind of way to look at um emergence but how do you find that point in in which there's the bi-directional information flow so this so that's that's maybe a better way to phrase my question yeah so i assign this great to discuss this with you but as i said i haven't done the whole research i am i'm happy to discuss my intuition here with with all of you but that's it yeah i so one of the hypotheses that i have is that you might find something like that welcome down downward conversation in biological systems i'm not sure of flocking behavior would be the place to find the workout session but i'm thinking about downward constant work station in terms of selection and it might be that you need some high order properties uh so that that sorry that high order properties explain some fitness and that in that way that calls somehow all the way down at the microscale at the micro level you observe some behavior as a result of the fitness that is uh sort of coarse grain or or or higher level so i'm thinking about that workout station in the in those terms but you might find other notions of downward causation there and i might found them in doing the research now in the flocking behavior i'm not sure but you might think that the the ability of many of many birds to to go in a particular to to do a particular pattern has some some fitness associated to it and if so that fitness which is fitness of a group might explain the individual behavior so if that's correct then you might have a notion of downward consensus that you don't find necessarily in physics but uh this is just pretty these are preliminary intuitions thank you blue so like going towards like along those lines um and going back maybe to the attractor so so when there is like a the emergence of a group does does that like reign all of the individuals in the group like closer to a new attractor state or closer together like tighter like a tighter cluster around an attractive state or is it like the new attractor state is an emergent property of the group or or how do those two fit together like in terms of fitness um and you know in biological fitness you have to be able to maintain this kind of non-equilibrium steady state density so in terms of fitness of a group or cluster and biological systems is that like a new attractor or is there yeah or just what are your intuitions maybe about that yeah so i i wouldn't i wouldn't try to describe the notion of of emergence here in terms of attractors but it might be useful in the end but i would like to uh to have a notion of emergence that is that doesn't rely necessarily on attractions because i think that finding attractors is is a hard task in biology and some of these models don't work with attractors anyway so i don't i don't i don't think the notion of emergence needs to be associated necessarily to attract maybe in some cases it would be useful to understand emergency in that sense but i wouldn't think that it's necessary to understand emergency incentives for attracting and i agree with you but but um like the coupling of emergence maybe to to me like i link fitness to an attractor and so it's it's hard for me to separate then if if fitness is linked to attractor but fitness is is not linked to um or is also linked to emergence then that that draws a line for me between emergence and detractor if that makes sense yes so that's why i will work with biologists on this project i'm not a biologist myself but i i i think that fitness can be defined also independent on the notion of attractors and i will try to take those notions cool we will try to adapt those definitions so dave just because you haven't asked a question yet then stephen yeah one of the reasons that uh looking for a track spending too much time looking for attractors associated with them with emergence is among the things that happens when you get emergent behavior is there's a more differentiated set of interfaces with the outer world and because of that there's less duplication and less resemblance among the subsystems already within the overall system cool thank you stephen yeah that's a good point around the challenge of identifying that attract especially if you've got this multi-scale nested i mean so you've got attractors that are only present for nanoseconds you know even if they're there and there's millions and billions of them you know you and then on top of that you've so i can imagine that it could be really useful and i i i'm curious um you see that you know the freezing of ice um and the way that it you know water becomes more dense and actually about two degrees then starts to get less dense and as it approaches freezing and then you sort of hit this phase transition um and i think there's a real use in that also being learned about for active inference because although an active influence doesn't happen directly in the field of anything to do with entropy there's this common misconception of neg entropy as if you actually get a negative form of entropy which is kind of in the in the kind of common usage but there's no such thing directly as a neg entropy if there's a there's a negative there's a change in entropy you know there can be a variation in entropic but it's not a stable state in the same way that you get with energy but that that um i think that can be a misconception sort of creeps in quite a lot but i'd be curious if um i you know um effectively you know there's there's there's a value of entropy that goes down spooky value of entropy which no one quite knows what's going on with the entropy we know there's a value for entropy that entropy goes changes in from ice to water but you know even richard feynman used to say well no one quite knows what's going on there and maybe this is something that you're trying to get into is what's going on in those phase and these entropic changes which doesn't have the benefit like you say of some a gentle driver of the free energy principle to make it happen so i don't know what your thoughts are but how that might even how that might be useful to help people who aren't as heavily into physics just think about the idea change and entropy because i think it's it's going to become of something that more and more people are using outside of physics if you have any dots otherwise just the maximum density is 4 c for water and it's super super slight but a very interesting pattern we can turn to well if anyone wants to raise their hand um or give a thought stephen you said like that the neg entropy as like it might not exist that was one thing and there's no such thing as entropy well you know yeah i also wanted to sorry about about that i yeah i also wanted to to stress that in equilibrium phase transitions in physics you don't use the notion of negative energy of course you just use entropy and standard statistical mechanics plus three normalization group methods so that my might be an asymmetry with some of the case studies that i analyzed in biology but yeah we'll see and one of the fundamental tradeoffs discussed in the paper um was this generality versus the plausibility and the realism do attractors even exist or is it just as if a system is moving towards something but does does something that is not realized exists and so this kind of uh takes us into talking about uh whether we're describing the system as it is or rather situating the system in a broader space of how it could exist and that was one of the fundamental points of the paper about the challenges of discovering the appropriate phase space for a biological it's just cool because these are knock down punches for fep and active nor are they trivial they're in that optimally informative or a confusing space you know between zero and one probability where it does matter and uh where philosophy comes into play where the the specifics of the situation come into play we can't just simply make arguments without reference to a model but then are we trapped to just describing specific models specific papers or are we able to go beyond that ever i'm just looking through um if anyone wants to raise their hand or any yeah about that sorry uh so we do rise a concern about the existence of attractors uh of course attractors don't exist in nature yeah because you need you need taking infinite limits in order to define attractors in a model and you cannot so infinites are not realistic right so we cannot take systems don't don't take an infinite amount or processes don't take anything out of time and uh you cannot repeat an iteration in renormalization group transformation infinitely many times in a in a realistic way of course these are all idealizations but in principle i wanted to to stress that we don't have we don't have a problem with that so of course any model that uses attractors wouldn't be realistic in a way so you will be using an idealization that's true but we don't have a problem with that if the idealization can be justified i think in many cases especially for example in in in the treatment of phase transitions in in physics where you use uh limits there's a way of justifying those limits and justifying the existence of attractors and therefore even if you miss some realism in the model of your of your physical system since that idealization is justified that that is plausible so that that model is plausible and it's legitimate what we are what we are worried about in the paper is that it might happen that sometimes uh the models that well they try to model many systems in terms of attractions are not realistic they cannot justify the idealizations that are being made in order to model many plenty of systems in so i just wanted to clarify that because we don't disagree in principle with the idea of of using models that appeal to attractors even if attractions don't exist in reality our problem is well with the possibility of finding attractors in the first place or finding asymptotic solutions and the second is about the possibility of justifying the idealizations that are involved when you use attractors or when you appeal to yes and like many other arguments in biology it's a little bit like the anthropic principle or fallacy depending on which side you're on it's like you mentioned in the paper you have to prove the existence of the attractor as well as as we discuss the time to conversions and then justify why certain denote homeostatic states and one biologist's answer is just those are the living systems if the attractor weren't a homeostatic state or a high fitness state that is not going to be an ant you see around for long and so we can just say that if there is an attractor and indeed it's as if there's an attractor for persistent systems that's the adaptive one period and the second that it's not that way the system fails to exist so that's the sort of circular but often impenetrable tautology of selection and persistence of systems so that's a nice uh transition perhaps into the question that i've had about um dynamic equilibrium versus a non-equilibrium steady state density so i don't know if you want to flip to that side daniel but um there there's you know several quotes in the paper where the i'm not sure exactly what's referred to um as a dynamic equilibrium i'm assuming that it's uh homeostasis but um i just was wondering why because a dynamic equilibrium is um a reversible chemical reaction and there's nothing about homeostasis that is i mean there may be some things but not everything about maintaining is has to do with reversibility um so i just was um wondering uh all biological systems actively maintain a dynamic equilibrium with their environment and what is meant by that this was a quote from the paper oh wait you're muted patricia but if you have any thoughts yeah so sorry i i was missing the screen somehow disappeared everything uh yeah blue so actually unfortunately matteo wanted to answer that in a in a in a neat way uh basically we took this from from freestone himself so he refers in many places and he had the quotations and so and i don't have that at the moment i can try to look at it but in a so you might be right it's not the right way of of describing the homostatic states but i think it's a mistake that that freestone himself does i think so this is basically what what matteo wanted to to say and he had these quotations in the papers where he does if you want to email him he will he will address you to those words but not no so in our paper depends on that notion actually so that might be a mistake that we make i think because we need to give freestyle but not so much depends on that so basically our criticisms have to do with the idea of modeling all biological states in terms of random dynamical systems in understanding homostatic states in terms of dynamical attractors as they are described in the paper we don't rely so much on the notion of dynamic equilibrium so that might be just a loose way of of understanding uh homostatic states so it's referring to homeostasis though not a dynamic testing okay yes yes yes tumustasis yes yes thank you stephen i mean this could be an interesting area for the for the ontological work that's happening in the lab because this question of you know is it dynamic equilibrium i think like blue was saying there can can be that kind of backwards and forwards of which there's maybe a bit more of an arrow in one direction and then you've maybe got the word equilibrium as opposed to non-equilibrium and then you've got this so he has this term dynamical systems you know and um as opposed to system dynamics which is the kind of the way that say chemistry like equilibrium work is more system dynamics for dynamical systems um and you've got this interesting thing because you've got this non-linear dynamical work happening at the sort of the the the physics level and this emerging property level and then at some point as it appears from the organism's perspective we might start turning and using the language of dynamic equilibrium ecologically you know or ecological and this this shifting the way that the ontologies are present which may be true because as things persist and interacted with as entities um they they appear to be a different type of dynamic but we are seeing from the active inference that that dynamic may in itself be more dynamical than we perceive so i don't know i just thought i'd mention um i don't know if we need an answer to or maybe daniel you've got a thought but there's something about this shift between biology and physics at the sort of at this sort of very um molecular stroke um you know theoretical level and then you've got the idea of what that means for biology and then you've got the idea of what that means for biological organisms and maybe this is something that we've talked about this pluralistic ontology that needs to be feasible in the field because like say it depends i think that was a very good point it does depend on where your models are based and grounded thanks mateo has some very interesting work on pluralism in biology and the philosophy of science so i'd recommend people look there now contingent explanations are not necessarily plural one can be an absolutist and say it depends on the particulars and there's only one correct story so it's an open question to what extent pluralism and particularity will coexist and agreed that at the high level as we mix metaphors take ideas across different fields we have to be aware of what is gained and lost what are the challenges and opportunities of applying terms from mathematics to physics and from physics to biology and taking us all the way up and down that xkcd comic that we were um looking at earlier question area patricia that i had was about action in the loop maybe there's two parts to it the first is just we often hear about fep as being akin to a principle of least action so just from a physics background what does it mean to be a principle of least action or what would it mean to be like a principle of least action and then um does it uh change the formalisms or not when we're integrating control of behavior kind of control theory formalisms or cybernetics into ensembl modeling because the actions of molecules are implicit in the phase transition of the ensemble of water but what happens when we actually have a control parameter in our equations like policy okay thank you so first of all i did find the papers where where frison uses the notion of dynamic equilibrium so for for blue and he uses a synonym of of homeostasis but i i cannot use the chat can you oh it's on the left side uh the bottom left it says that enter a nickname to use the chat you can pick up whatever you want it was kind of possible okay call me trim tab okay it was so i thought it was it was sort of a password so you just wanted to allow everyone to write in the chat it doesn't require a child name but good these are the papers blue so you can take a look at this so yeah but as i said not so not so much in our paper it depends on that notion of that so you might disagree with that lack of record there but yeah and don't blame us for that we just okay nice that was it interestingly not in a peer review paper not that it explains everything but he may be speaking maybe yeah informally in this aon piece exactly i think i think it's just uh yeah a more loose way of talking about it and we also use it because it's useful then when you speak about dynamical attractors so if you have a notion of dynamic equilibrium it's just uh useful but of course we were we weren't thinking about the the the the proper definition of dynamical so good now just return the action question yeah okay okay so yeah again so we are here so sorry that i excuse myself of summarizing the work that frison did uh but here we are just summarizing so we don't give a proper definition of principle of list action we're just relying on on friston's own definition and he basically in that paper of 2012 he needs the principle of of least action in order to formulate the free energy principles so the principle of least action is a step an intermediate step in order to formulate the free energy principle nothing more then i would be more careful in talking about principles of list action in in the fa in phase transitions in physics i think you might not even require any principle of list action there you might talk about those uh ground states as principle as uh sort relying on a principle of less action but i would be more careful about uh exporting that notion to to the physics of phase transition what's that kept up yes so that's the first part about the principle of least action and then just is there um something that is changed markedly when we introduce these control theoretic formalisms like policy selection into physical physically inspired or based models that entail activity like the thermal motion of particles but are not approached from a control theoretic or behavioral regulatory perspective are you thinking about social systems for example perhaps different kinds of systems but um mainly just a uh a particular system uh in the way that uh free energy principle for a particular physics for students 2019 work like the particle is the blanket and the internal states that's engaging in active inference let's say and so it is implementing inference on policy it's selecting policy it's behaving and so is that sort of policy selection latent within non-controlled theoretic activity-based physics models again like the thermal motion of molecules or when we introduce policy does that change some of the qualities of the system i i think so it's an interesting question i would need to think about that a about it more carefully but my intuition is that it does it changes the the way uh i still understand uh phase transitions in physics i think that requires changing formalism and again it requires uh important well making important difference in your in your model with respect to to how you model phase transitions in physics i agree something changes and it's not for linear time discussion necessarily to unpack but these are the kinds of questions that we're getting at absolutely so blue yes yes i think so um i just want to reference the um articles that you posted in in the chat so the am thing they did say dynamic equilibrium but like i like don't trust like op-ed science writers to make direct uh correct quotations um and in the ramstead paper that was cited in your paper the markov blankets of life it specifies that dynamic equilibrium specifically is not the state of homeostasis it says the very existence of living systems can therefore be construed as a process of boundary conservation where the boundary of a system is its markov blanket that means that the dependencies induced by the presence of a markov blanket are what keep the system far removed from thermodynamical equilibrium not to be confused with dynamic equilibrium so i mean specifically like there it's not dynamic equilibrium it's it's far from thermodynamic equilibrium but but still dynamic equilibrium is is not not a part of that and so that's why that reference was very um confusing to me and like they on paper is like yeah like yeah and then we made the same mistake that that he made without correcting it in the way that it can be so i i agree we were talking loosely there and we were what we were having in mind was the notion of dynamical attractors so that's why we found uh dynamic equilibrium very very like a a nice term but but we might be talking loosely there but as i said so that might be a mistake but um not so much in our paper depends on that notion well so and the notion of a tractor in in the state of homeostasis is specifically a far from equilibrium removed space that a biological system exists in and so it's just kind of um yeah confusing to me i thought interesting um you know it's a challenge to say enough to identify what we're talking about but especially when we're crossing disciplinary boundaries to know which pieces of baggage make it through one border or another like dynamical could merely just mean involving time it also brings on a lot more meaning in different areas it's a great yes so we talk we talk in the paper about the possibility of uh attractors like that change well one attractor following another so we understood that as a loosely as a notion of dynamic equilibrium but you are not using the term rigorously probably here let's turn in the um sort of home stretch of however long we want to speak to the short and concise conclusion you you in the last paragraph brought up this really interesting point about the fep as a definition for a system and so um i wondered what else could it be or might be what might we want a framework to be other than a definition like what else in the counter factuals of the way that this paragraph could have been written what else could the fep or another framework be other than a definition is that synonymous or complementary with a description what other pieces come into play here can you can you repeat the question i don't know if i understood it um if the fep is not a definition what else could it be or other than a definition what do we want for a full pledged model principle what would make it a principle i think something that can be can explain life or it can explain um yeah why why why uh biological systems uh apparently violate the second law why do they stay in that far from equilibrium steady state i think uh many defenders or advocates of the applicators of the free energy principle understand this sort of as a principle that explained that has explanatory power that explains the the the not only the existence even the existence of living systems but especially why those living systems stay far from from equilibrium or from from maxwell boltzmann equilibrium for a long time in a in a steady state great point so yeah that's smart and explanation which of course goes beyond or is complementary to definition explanation is one of the key terms and themes in the history of the since an explanation is evaluated subjectively or inter-subjectively like was that an adequate explanation that's a statement about the relationship between a perceiver and the explanation itself and so it uh dumps the entire mess back into the lap of almost a psycho-historical perspective on science because good explanation is about the enculturation of the person evaluating how good that explanation is so steven and then blue yeah i was just um looking up regarding dynamical and dynamics so dynamical the adjective means relating to the study of dynamics so it says a dynamic system is a system exhibiting continual change so a dynamic system is continual change a dynamical system is a system relating to the study of the dynamics so a dynamical system is like a system working with the dynamics of a system and dynamic itself just talks about the actual fact that there's change happening so that's kind of um that's kind of interesting and be interesting as well whether you get dynamical things or there has to be systems you know like the the is there that may be where there's some sort of interesting play between uh phase transitions as well like maybe there's some there's a dynamic going on is there a dynamic that is second order to the first dynamic in some way so that that's kind of interesting yes things and processes and static versus dynamic and then loose but acceptable in many cases use of language again the text modeling will be superlative when we can establish where these adjectives and nouns and verbs are being used appropriately versus inappropriately um or at the very least suggesting more precision or less precision than uh warranted so blue oh and dave has a great comment there historical examined by historians using founded methods versus historic which is like about the event itself like it was a historic victory or something like that so it's uh interesting how we give the disciplinary nod with that al on the adjective um i was just thinking about the difference like linguistically between historic and hysteric like what might what might that be um so i my question unrelated to historics and hysterics um really goes back to um you know what patricia was saying about the second law and touches back to what stephen was saying earlier about um the non-existence of negative entropy and and um you know i'm i'm curious about that because i think um a lot of the fep is is um grounded in information dynamics which depends greatly on the existence of negative entropy like the minimization of surprise uh in the terms of information theory so i i was wondering if physics and information theory are are different are they one and the same what are the differences how do we clarify them and similarly um you know one of the claims in the paper is to pay attention to the differences between physics and biology and what you know i've always thought the laws of physics are universally applicable and and perhaps they're not but but um i i think that they should be and you know that this defiance of the second law like are do the laws of physics not apply in biology in biological systems um is is it negative entropy that that biological systems do i mean scott david always classically refers to the entropy secretion of you know biological systems and so i just was wondering how do we reconcile these differences maybe if you guys have thoughts yeah great questions stephen you want to go on that topic or yeah i think i think there's a really i think this is important because you see neg entropy kind of is a bit like negative enthalpy in the way it's formulated i but the thing is you have a reduction only relative to a starting state starting equilibrium state in terms of entropy it's it might have gone down in terms of a starting state but it's not actually lower in like energy is energy right you it can't be made can't be lost you know but the entropy you can have a variation in the entropy and that variation can be noisy going up and down you can then extract um you can have things which are more ordered now that's fair enough you can have more order but neg entropy um isn't something which is a value in that way it's just a convenient piece of temporary mathematics to help see whether you're going to have a force driving or reducing a reaction which so that's where the challenge is because it's you're getting it from the variations not from something which is an equilibrium state because as soon as you say neg entropy it makes it sound like it's sitting there with a particular amount of that makes sense um and it's not really like that so this this is this is like one of the misunderstandings i think around entropy okay interesting um one other view blue on your question about information and thermodynamics we have some very slam dunk info thermo on the processor scale like the amount of heat that is emitted or absorbed by certain very very granular atomic in multiple senses informational transitions and so it's almost like that's like the the stem and there's that we know that there is a connection there um either to be sure if not in actuality so map and territory information and thermodynamics potentially coming together and part of our curiosity and drive with active is whether info dynamics might apply to higher orders of organization and whether we can use some of these models and ideas to describe informational patterns and thermodynamic eventually even bioenergetic patterns and phenomena beyond the cpu and that's where it's going to be this question of okay maybe we generalize and it's useful but we've lost some reality touch so the a temperature parameter on decision making doesn't refer to the thermometer reading of the brain i think we're okay with that so in a way we are lifting away from the physical grounding because we're using terms like temperature in their information and not their physical capacity and as the key questions raised in this paper um brought to this discussion there are opportunities and challenges with making those kinds of generalizations it's not just a simple win it's things are gained things are lost some things come into resolution other things become incapable of being resolved so that's like the some of the excellent questions i think the paper brings to the table and it combines those questions along with just tantalizingly a look at the history and the development of active inference as a framework and so we can be clear about which papers said what when and then where are we headed what would be our preferences for a thermodynamic model of decision making will we only succeed when it's about the temperature of the brain it might be for some person however other people might just say if there is a temperature-like parameter in the sense that that parameter's formalism looks like temperature of water in a test tube then we have something that's already useful and interesting that's where the train ends for me and then somebody else may continue to build the rails off towards wondering whether is it metaphysically a temperature of decision making all these other areas that people can take it and so it's this trade-off between realism for generality and utility that we've come to these uh meta modeling discussions before in number 14 and beyond anyone wants to ask a question in the chat or raise their hand here otherwise let's see back to our dot one questions and see if there's any threads from dot one that we want to just um recap or see if we have a different perspective on and then we can just sort of close the dot 2 and jump off into the next steps so we started with some questions again from a more history of fep and active in what ways has physics been used to investigate the fep we talked a little bit about how the decosta paper that was 26 bayesian mechanics for stationary processes relates to the erogedicity stationarity or not we'll see we talked about then and also a little bit today is the fep a theory of life well in the conclusion it was said to be a definition and we talked about some other things it could be blue so you still have to unmute you'd think i i would learn after all this time um never learn don't change the um the ergonomy thing i think maybe you guys discussed that uh before i got here but but i think like one of the the points that was made in this paper um addressing the ergodicity and the fact you know biological systems are are i mean i don't think can i don't think biological systems can be described as as ergotic and this is a discussion that that daniel and i have had many times um i mean sure within a very restricted space like you know my body temperature for example in the range from i mean what can i survive like 105 fahrenheit probably and you know maybe 95 fahrenheit so i have like this 10 degree range and i'm sure at some point in my lifetime i'll visit all the space within that range because you can't skip over you don't go from 96 to 100 and skip any of the points in between so like you know it's incrementally like of course i'll visit all of that space um and so i i do think that that that's an important question but um yeah that was really um addressed by de costa in in that uh 26 live stream um about you know homeostasis is a quasi stationary because it's not perfectly stationary right it's like a it's a squishy stationary but it's not you know it doesn't it doesn't sample the entire we don't sample the entire state space of say temperature um you know as a biological system that must maintain a certain temperature so i think that that was resolved but did you guys discuss that before i got here not the boundedness per se potentially some relevant pieces about arrogancy and stationarity um yeah so may i say something sorry though yeah i have to turn off the video i'm with my baby here hello baby um yeah um yeah so we didn't we i agree with you blue i think that's one of the things that we that we discussed in the paper so it's very problematic to understand biological systems such as body systems uh but uh in the discussion last week you were saying or maxwell was saying new formulations of the free energy principle don't rely on that assumptions that they dropped that assumption in the new and what i was saying before today is that it's still in new they rely on the existence of weakly mixing random dynamical systems and to my understanding mixing systems require ergodicity anyhow so it's not clear that they have really dropped the assumption in the new formulations and if so i would like to see exactly how the framework can work without the assumption of erodicity i haven't seen that explicitly i my belief that that can be done but i would like to to see it we don't evade requirements just by not mentioning them so it's about what papers do mention as well as reading between lines between the formalisms between the citations to really what is the bedrock or tautology that the theories rest upon stephen you know i think this also might point to your um cybernetics point daniel um in the sense of you know where does something happen at this kind of ergodic process of mixing and where may it happen through a cybernetic mechanism that the organism has that is enabled by sort of active inference or whatever processes emerging but the fact that it's emerged from ergodic processes at multiple nested levels doesn't necessarily mean that some of the higher level ways of keeping something within bounds then has to follow that because it it you know you can create a out of all the cells that are doing their thing so to speak or being their thing so that we we this is where it can maybe be that danger that yeah there's an over extension of a certain process it's like okay where does something which is relying on ergodic processes some other approaches able to give way to another type of um control theory yes and over short time horizons is it possible to have sensory motor loops that work whether or not some infinite case is what it rests upon blue so patricia i'm curious um to know or to know what you think about if you've read the decosta paper about um basinary basinary processes for bayesian process bayesian mechanics for stationary processes there you go um so i'm curious to know if you've read that and what you felt was perhaps lacking from that mathematical description to um meet the requirements that uh you were referring to for mixing systems and ergodicity yeah thank you i i did look at that but we explicitly wanted to focus on other assumptions not on the bayesian assumptions of the paper we didn't also mention the markov blanket assumption which is a core assumption in and we left that out deliberately first of all because of lack of space the paper is long enough second because there is much more literature on that than on the other these other assumptions that we discuss in our paper so i think in this formulation of um well i don't i don't have that paper on the top of my head but i think they they again uh rely on weekly mixing dynamical systems in the paper right they don't we can check they they use that framework so i think to the extent that they use that framework if i'm not mistaken then the same criticisms apply you need to find phase spaces and you need the existence of attractors at least cool i just imagine like these networks of citations like a some sort of biological metaphor some swarm of citations with words being traced between them and connections of the words and um i think um we're in the boutique the artisan the farmer's market phase of these kinds of analyses and there there will be a time not so far away when these questions can be resolved better um so yeah just like the paper aside i mean i i definitely realize that there are space constraints and also like what was the timing of your paper versus the decosta paper and did you have time to like incorporate all of those the new like updated um version of the maths uh so there's not weekly mixing or mixing mentioned in the de costa paper and so i really just would be curious maybe you want to follow up via email um to to know what what was missing from that that formulation in in your mind for it to be um plausible because for me too it was very uh like the the foundation of the fep existing in in ergodicity or requiring ergodicity for applicability was very um jarring like i felt like that's that can't be right but um i i felt that it was really resolved by the decosta paper and so i'm curious if you um don't feel the same way what your hesitation would be so maybe we want to follow up in in in a dot three at some point yeah or maybe or maybe by email so i would definitely look at this paper and other new research because that was published this year right it's very recent so the papers that we were focusing a little bit more on in the babe in our paper was par it is also very new and christian 2019 this prop this free energy principle for particle in the other papers we look at them uh very briefly because they were actually out after our paper was accepted or maybe during the review process so it wasn't part of our research systematic research but we did take a look at them uh i think we might have even quote that but yeah so i i would like to take a look at that again i'm not sure whether they use if they use the notion of information geometry then they probably also rely on weekly dynamical systems but i i would have to look at that again so we can continue the conversation by image counter factual paper spaces you know you have some sort of clause that gets triggered if something happens in the literature this paragraph should be deleted like pending a future word on this this could be deleted in the last little bit maybe we can just have any thoughts uh just why does this whole discussion matter for the fep and for active well the validity and the adequacy were how it was described by this paper it it matters so it's kind of cool because often the citations that we discuss they don't always lay out on the table why it matters to understand this topic and uh i think we got that very clearly from this work so i really appreciated that and it also laid out many of the next steps directly not every next step but the specific key uh themes and challenges are addressed and if there is a obvious or non-obvious response that'll be a citation that somebody can swoop on stephen yeah i think it's important um because we are seeing a lot of a huge amount of interest from in activism and embodied work and physicality and how these things can shape up and therefore purely mathematical arguments will limit things i think getting into the nitty-gritty of the physics which in itself is wrestling with some of this because it's kind of cutting edge for them in terms of that is really important though i think because you see some of the criticisms coming around in activism um trying to get arrested on things and i do think i will say that i think maybe sometimes saying go and visit the math which is ultimately based on ergobicity which is basically a physics principle as much as a math principle in terms of you know how that comes to be present comes up so i think this is this is a really important area that physics and and maybe the sort of the areas of applied physics start to come into yes and we'll just send them on the runaround okay go to the math and they'll go to the physics okay now to the philosophers you go and then back to another maybe non-academic department is that run around the answer what if there's a stationarity in the runaround very good points um it's kind of fun because we've had these questions for a long time if not the longest time yet many of them were explicitly addressed by this paper and by the 31 discussions so that's kind of cool to see like a sort of it's not a conclusion on our series but these these uh questions which are often the next step for other papers discussed in the dot 2 discussed in the conclusion or the discussion that's where this work picked up and so that that's a um a different attribute than we've seen from some of the other literature we discussed dave do you have any other thoughts patricia would you like to give any final comments i think i have said enough unless you have an uh a last question what papers i i'm a little bit of fun i'm sorry i'm a little bit of an outsider in the free energy principle i must confess but um i i would like to see how it progressed so as i as i said at the beginning we are not against the the free energy principle a priori so if our paper can lead to more research in the free energy principle direction we would be happy and we would consider it a success so we are not necessarily enemies of the free energy principle and i'm in principle very statistical mechanics and even the language of dynamical systems to biology so i am in principle interested in this type of of research and also in the active inferences interdisciplinary so i would be very interested in see how this evolved is and if we as philosophers can contribute to that by criticizing some points that we find non-rigorous enough then our job is done so that's basically our goal thanks patricia totally agree this um hopefully we can enact the kind of dialogue at the micro like a conversation and a disciplinary or scholarly dialogue that plays out over years the philosophers have a very important role in that blue so what uh patricia was just saying reminds me of um you know when we discussed mike levin's paper in the summer um the computational boundary of a self it's it's you know there's the instrumentalist view and then the realist view and then the utility axis in that in that uh diagram and so definitely um i thought that this paper was useful in provoking dialogue and um really kind of uh i would like to leverage your vantage point to kind of grind into some of the deeper maybe more mathematical stuff that's kind of beyond me because it's not really my my background but to kind of you know dig into that hole and make sure that there's a solid foundation under the hole or is it a hole that pokes all the way through and leads to nothing on the other end so i think um you know it's useful for for getting at those those questions my last question patricia would be what areas of literature or papers might you recommend that we dive into so that we can have a comprehensive view know where we're coming from and where we're going i have i can mention many many babies so you can do that better by email again so if you're interested in literature and some specific research or topic just send me an email and i can recommend philosophically to you but i would like to especially recommend one paper so if i can choose and i don't know the title right hopefully but it's it's a paper by karim thibault uh who is called something about imperialism and migration so i can look at that now because it really it's a very general paper in philosophy where drastic they draw a very nice distinction between migration of models and imperialism of models that i think will be also useful in the interdisciplinary research that you are doing uh because they they stress the difference about using the methods for or exporting the methods from one discipline to the other in a non um imperialist way which would imply that in the new discipline you have to justify the model you have to justify the the the mathematical methods that's that's the one models on the move migration and parallelism brilliant and they distinguish between that that they call migration of models from imperial imperialism in which you just extrapolate the same models from one discipline to the other and you try to expand your discipline which would be sort of the the attitude that you were you were um that was illustrated in the in the comics that you were discussing last last week so i think that paper is especially useful for for for people doing interdisciplinary research because it's a very useful distinction this model is on the move thanks for the awesome suggestion so just to the point about emails or however it goes just consider the dialogue opened we really appreciate that both of you came to discuss and also interacted with the dot zero and dot one videos so just awesome work with uh research and thanks again for helping our lab do what it does and this is also an excellent suggestion because it's a provocative title but we're talking about provocative ideas and with broad scope and this kind of takes that map territory discussion to a new place potentially talking about movements and imperialism and colonization all these other things that come up in different terms but we've noted this in our upcoming streams calendar so after our lab meeting next week we'll have stochastic chaos and markov blankets so that will complement some of the things that we didn't talk about in 31. we'll talk in 33 thinking like a state with aval aka serval and then we have two more slots open so perhaps we'll select one of these that we've already listed or perhaps somebody wants to join if an author wants to discuss other than that i think that brings us to a close today so again big appreciation to the authors for their work and engagement with the lab and thanks steven dave and blue and everyone else who participated in maxwell for joining the dot one so good times and we will see you next time thank you thank you very much bye